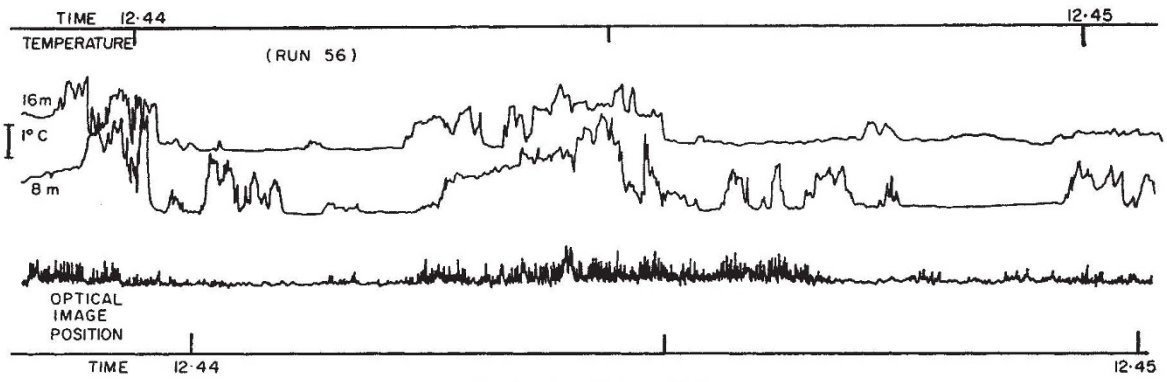


Fig. 1. Hay, 15 March 1965.

gests the basic importance of wind direction in the solar seeing problem[7]. In addition, there was then found to be little or no correlation between occurrences of optical and temperature quiescence; presumably this arises from the experimental arrangement with temperature sensors at one end of the optical path, so that with a cross wind the optical and temperature measurements are affected by different parallel air streams.

Thus in summary we can say that current micrometeorological knowledge has suggested the conditions (sunny, with sufficiently light wind and suitable wind direction) in which intervals of good seeing along a slant path are likely to be encountered. This experiment has given a direct demonstration of the occurrence of intervals of good seeing, in association with the temperature quiescent patches of air, under these conditions. It remains for more comprehensive investigations to establish the meteorological/optical relationship in greater detail.

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Palaeomagnetic Investigation of an Old Red Sandstone Formation of Southern Norway

Palaeomagnetic measurements of Devonian rocks of Europe have posed an intricate problem, because the red sandstones from this period, which spread from Wales across Central Europe to Siberia, exhibit palaeomagnetic poles significantly different from that deduced from the Lower Devonian lavas in the Midland Valley of Scotland[1]. Recent thermal demagnetization studies on some Old Red sandstones of the Anglo-Welsh cuvette[2], however, have revealed a weak component of high thermal stability with a direction of magnetization which is in good agreement with that found in these Lower Devonian lavas. Furthermore, application of the fold test has shown that the magnetism of this high temperature direction is Pre-Hercynian. This result, together with the apparently high magnetic stability of the Scottish lavas, led Chamalaun and Creer[2] to suggest that the lavas give the best estimate of the direction of the Devonian field for Great Britain.

The Devonian red sandstones of Europe and Russia which have so far been collected for palaeomagnetic purposes unfortunately come from regions which, to a greater or lesser extent, became geosynclinal areas in subsequent geological epochs. It is possible, therefore, to suggest that at least partial remagnetization, induced by moderate heating during burial as proposed for the magnetization of the red beds of the Anglo-Welsh cuvette[3], may have occurred at a later time over quite a large area.

Since the end of the Caledonian orogeny most of the present Scandinavian peninsula seems to have been a stable continental block, and negligible sedimentation appears to have occurred in the post-Devonian history of this area. The palaeomagnetism of Devonian rocks of Norway could well be of great importance because it may provide a means of verifying the theory of remagnetization.

The palaeomagnetic data presented in this communication are based on samples of red sandstones collected from the Røragen Devonian, a complex covering an area of approximately 6×2 km. The area was mapped in 1913 by V. M. Goldschmidt[4]. A simplified copy of his map is shown in Fig. 1. Floral evidence[5] from the lower part of the group has fixed the age as Lower Devonian.

The Røragen Devonian is built up of continental clastic deposits, ranging from shale to conglomerates. The conglomerates represent the major rock unit and are particularly abundant in the upper part of the complex. Typical red beds contribute very little to the lithology as they are represented locally only as intercalated layers of thicknesses of about 1 metre. The maximum aggregate thickness of the sedimentary complex may be 700–800 m and the whole complex shows a general dip of about 50° towards south-east. Locally, however, the attitude of the layers may be much steeper. To explain the dislocation of the complex, Goldschmidt postulated faults along the south-eastern margin resulting in a general dip towards this direction. The area was re-examined recently by P. Holmsen[6], who found evidence of the whole complex having been folded; he rejects the hypothesis based on faulting.

From the geological observations available[4,6,7] the evolution of the Røragen Devonian seems to have been as follows:

(1) Denudation products of the Caledonides were brought into a subsiding basin during late Caledonian times, but the occurrence of alternating layers of sandstones and conglomerates (containing boulders of varied petrographic composition) suggests orogenic movements also during deposition causing uplift of the pre-Devonian surroundings.

(2) After consolidation of the sediments, the complex was emplaced by folding and possibly also in part by thrusting (probably of very little importance for the

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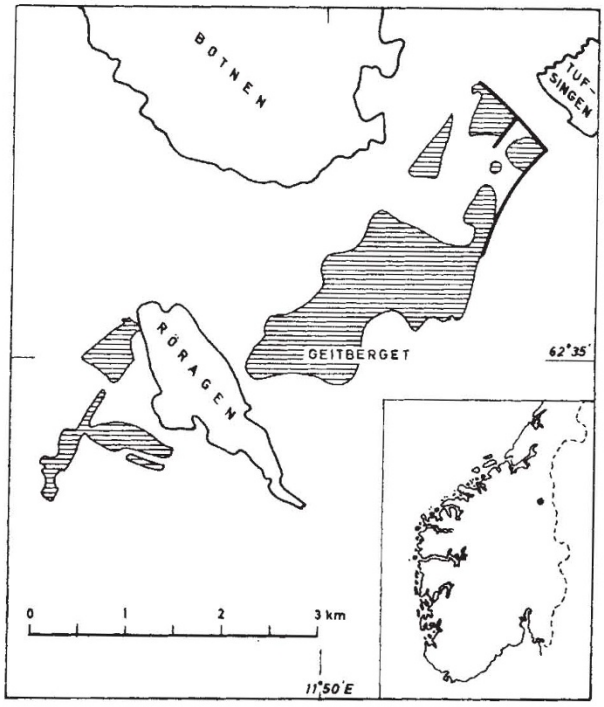


Fig. 1. Sketch map of the Røragen Devonian (shaded). Along the north-eastern margin of the complex are shown the visible parts of Goldschmidt's (ref. 4) postulated faults.

present study) during a late phase of the Caledonian orogeny.

So far, sixteen oriented samples from eight different layers have been collected at or near the Geitberget hillock where most red beds are located. Samples were collected over a stratigraphical thickness of at least 75 m. About five rock cylinders (19 × 19 mm) were prepared from each block and the directions and intensities of any remanent magnetism were measured on a sensitive astatic magnetometer. The intensity of magnetization was in the range 1×10^{-6}–2×10^{-4} e.m.u./c.c., averaging about 6×10^{-5} e.m.u./c.c. and the direction was consistent in nearly all the samples. Cylinders from all samples were subjected to a step by step thermal demagnetization. Fifteen cylinders were heated in air during each run and were rapidly cooled in air to room temperature after each step in the absence of a magnetic field. Except for a few weakly magnetized samples, which possessed scattered directions of natural remanent magnetism, the directions of magnetization are stable against the thermal treatment. Three samples, two of which have an intensity of the order of 10^{-4} e.m.u./c.c., seem to contain two magnetic components. This is suggested because, in addition to the Curie point of haematite, at which the remanence disappears, there is substantial decay of intensity between 500° C and 550° C, which suggests that magnetite is present. In all other samples, haematite appears to be the only carrier of the remanent magnetism and in many cases there are only negligible reductions of intensity when demagnetization is effected at temperatures below 650° C.

In a few weakly magnetized samples the directions begin to scatter after heat treatment above 400° C, probably as a result of induced moments caused by small residual fields inside the furnace. Because the total artificial thermoremanent magnetism imposed on some specimens is of the order of 10^{-3} e.m.u./c.c., small residual fields may introduce stray moments of significant magnitude in weakly magnetized material.

Two samples from slightly different stratigraphic levels in the same layer were magnetized in opposite directions. Because the remanence in both cases appears to be carried entirely by pure haematite it seems reasonable to assume that the difference in polarity reflects a field reversal and thus indicates that the remanent magnetism was acquired at the time of deposition.

The directions of magnetization as shown in Fig. 2 are given with respect to the bedding plane, which at the time of formation (and magnetization) was probably more or less the "palaeohorizontal". The overall mean direction irrespective of sign—assigning unit weight to each sample—is N. 210° E., +9°, and the radius in the circle of confidence at the 95 per cent probability level[8] is 15°.

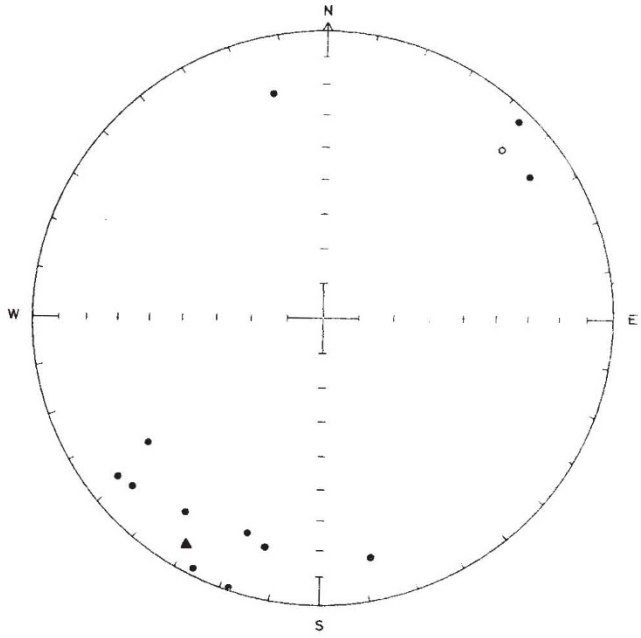


Fig. 2. Equal area projection showing sample mean directions of natural remanent magnetism of the Røragen red sandstones. Full circles are plots of north-seeking poles in lower hemisphere and the open circle is a plot in the upper hemisphere. The triangle indicates the overall mean direction.

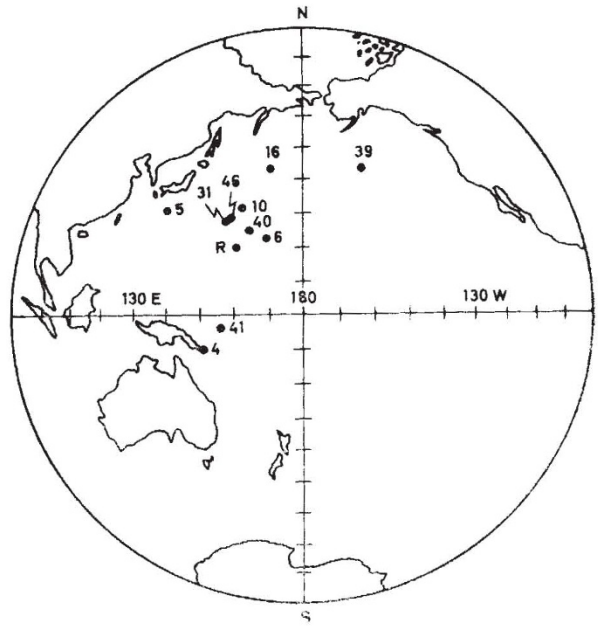


Fig. 3. Devonian palaeomagnetic poles from Europe and northern Asia. Numbers are according to Irving (ref. 9) and *R* denotes the pole position deduced from the Røragen red beds.

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From the mean direction and the assumed geocentric axial dipole field a virtual pole location at the present geographical co-ordinates 160° E., 19° N. is derived. This pole position is in good agreement with the red sandstone result from elsewhere in Europe and Russia and significantly different from that obtained by Stubbs[1] based on the lavas of the Midland Valley of Scotland. The magnetization of red bed formations of Central Europe and Russia therefore seems not to be satisfactorily explained by remagnetization, and the possibility still exists of there having been two Devonian field directions.

This research has in part been supported by "L. Meltzers Høyskolefond", "Nansenfondet og de dermed forbundne fond" and by the Norwegian Research Council for Science and the Humanities. We thank Mr. E. Halvorsen for carrying out the thermal tests.

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Strontium-90 and Caesium-137 in Precipitation in Finland

Precipitation has been collected monthly at seventeen stations in Finland since 1961. Until the end of 1963 the rain-water samples were collected with stainless steel funnels which drained into ion-exchange columns. The columns were placed in boxes fitted with a thermostat, and in winter the funnels were heated. Comparison of the results obtained from samples collected by ion-exchange and from those collected directly into polythene bottles, however, indicated that the former method of sampling does not work satisfactorily in cold regions. In poorly stabilized conditions, caesium, especially, seemed to run through the resin. For this reason, dry and wet deposits have been collected since 1964 with the funnels draining directly into polythene bottles.

Radiochemical analyses of strontium-90 and caesium-137 were carried out according to the method developed by Osmond *et al.*[1]. Fig. 1 shows the monthly averages for the deposition of strontium-90 and caesium-137 and the precipitation at the seventeen stations.

Deposition of strontium-90 and caesium-137 at the stations was about 40 per cent of that of the preceding year, while the amount of precipitation was about 15 per cent higher in 1965 than in 1964. The values found for the deposition in Finland agree well with those found in other places between 60° N. and 70° N. in similar weather conditions[2,3].

I thank Prof. K. E. Salimäki for his advice during this work.

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Amino-acids in the Mountsorrel Bitumen, Leicestershire

The Mountsorrel granodiorite[1-3] is situated 7 miles north of Leicester. The intrusion is boss-like in form, about a mile in diameter, and is considered to be Caledonian as it has an isotopic age of 403 ± 18 m.y.[4,5]. The mineralization which occurred within the granodiorite after consolidation has been studied by many workers; the most recent is King[6], who subdivided the mineralization into five stages, the first two being pneumatolytic and the others hydrothermal.

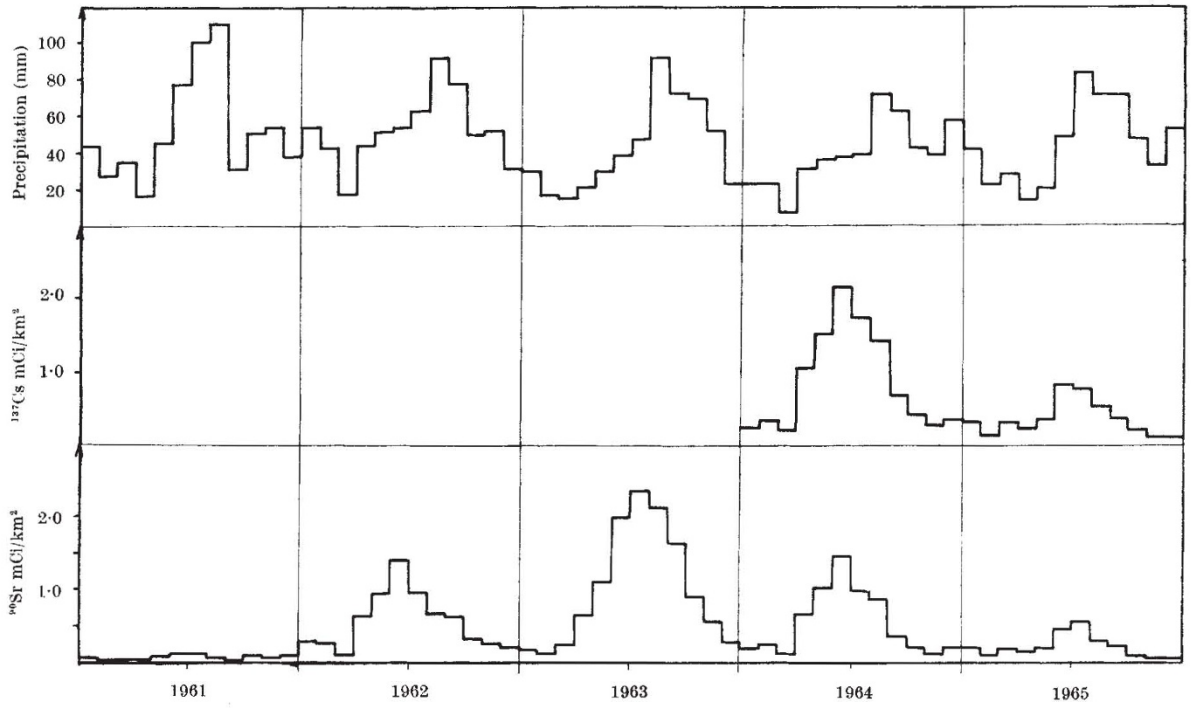


Fig. 1. Deposition of strontium-90 and caesium-137 and the amount of precipitation at the seventeen stations in Finland.

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